

1. Dataset Overview

- ❖ This dataset contains 72 air quality observations for Lucknow collected from all 6 CPCB monitoring stations on a gap of 10 to 15 days.
- ❖ The dataset includes Date, City, Recorded Station, AQI, Prominent Pollutant, and AQI Category.
- ❖ The objective was to clean and analyze the dataset to identify important air quality trends.

2. Cleaning Steps

The dataset was inspected for missing and inconsistent values.

- Four AQI values, four AQI category values and one prominent pollutant was missing.
- ❖ Missing AQI values were replaced using the mean AQI,
- ❖ while AQI categories were recalculated according to CPCB AQI classification standards.
- ❖ One missing prominent pollutant value was replaced with the most frequently occurring pollutant, PM2.5.
- ❖ No duplicate records were found. Date formats were standardized for consistency.

3. Key Patterns Identified

- ❖ The analysis showed that the Moderate AQI category occurred most frequently.
- ❖ PM2.5 was the dominant pollutant in the majority of observations, indicating that fine particulate matter is a major contributor to air pollution in Lucknow.
- ❖ Most AQI values fell within the Satisfactory to Moderate range, suggesting moderate air quality conditions during the observed period.

4. Anomalies Observed

- ❖ One observation was classified as Very Poor, indicating an unusual pollution spike.
- ❖ Missing AQI and pollutant values were also identified, highlighting minor data completeness issues.

This cleaned dataset is now suitable for further visualization and environmental analysis.

Thank You